Bulletin of the National Research Centre

RESEARCH Open Access

# Monthly fluctuations in total polyphenol content of Sri Lankan tea cultivars in relation to climatic factors and drying techniques

K. G. Nelum P. Piyasena[1,2*], Jayani Kalinga[2], W. S. Jayasinghe[1] and M. A. B. Ranatunga[1]

## Abstract

**Background** Diverse and variable climatic conditions of Sri Lanka significantly influence the biochemical properties of tea leaves. Therefore, tea manufacturers regulate oxidation/aeration durations during processing to optimize black tea quality and market value. The sensory qualities and health benefits of tea are primarily attributed to its polyphenolic compounds. This study aimed to evaluate monthly variations in total polyphenolic content of black tea and tea leaves over two years, offering insights into seasonal fluctuations. Analyses were conducted on black tea (BT), oven-dried (OD, at 60 °C), and freeze-dried (FD) tea leaves from two widely cultivated Sri Lankan tea cultivars, TRI 2025 and DT-1.

**Results** A distinct seasonal variation in total phenolic content (TPC) was observed in both black tea and tea leaves. For TRI 2025, average dry-season TPC was 25.03% (FD), 24.76% (OD), and 21.27% (BT); for DT-1, values were 24.60% (FD), 23.97% (OD), and 20.30% (BT). In contrast, wet-season TPC levels declined: TRI 2025 showed 22.99% (FD), 22.85% (OD), and 18.59% (BT), while DT-1 recorded 23.84% (FD), 21.77% (OD), and 16.82% (BT).

**Conclusions** TPC in tea is strongly affected by seasonal climate, processing method, and tea cultivars. Dry months consistently produced higher TPC in both fresh leaves and processed teas, whereas wet months were associated with the lowest levels. Among processing methods, freeze-drying retained the most TPC, followed by oven drying at 60 °C, while black tea processing resulted in the lowest TPC retention.

**Keywords** Black tea, Freeze-dried tea leaves, Monthly variations, Oven dried tea leaves, Total polyphenol content

## Introduction

*Camellia sinensis*, commonly known as the tea plant, is one of the world's most widely consumed beverages (Nelum et al. 2023). Tea leaves are remarkably rich in total polyphenolic contents (TPC), which are present in higher concentrations compared to other metabolites such as pyridine alkaloids, carotenoids, minerals, organic acids, and vitamins (Basnayake et al. 2025). The presence of these bioactive compounds is considered a key factor in the widespread global appeal and consumption of tea (Piyasena et al. 2024). Tea consumption is linked with various health-promoting effects, such as anti-aging,

*Correspondence:
K. G. Nelum P. Piyasena
nelum.pi@nifs.ac.lk
[1]Biochemistry Division, Tea Research Institute of Sri Lanka, Talawakelle, Sri Lanka
[2]National Institute of Fundamental Studies, Hantana Road, Kandy, Sri Lanka

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

anti-inflammatory, and anti-obesity properties. Additionally, it has been linked to a reduced risk of chronic diseases and may contribute to improved digestive health, immune function, and oral hygiene (Piyasena 2025; Piyasena et al. 2024; Piyasena et al. 2025). The TPC in tea leaves, which also determines the sensory quality of tea, is largely responsible for these therapeutic benefits.

High TPC in fresh tea leaves is a key indicator of the quality of black tea, contributing to enhanced flavor, and color, whereas low TPC is frequently associated with reduced quality in black tea (Aaqil et al. 2023). Furthermore, it is reported that TPC of tea is highly sensitive to seasonal variation and tea processing methods (Ahmed et al. 2019). The chemical composition of tea leaves is influenced by a wide range of factors, including genetic background, cultivation region and altitude, agronomic practices, harvest season, leaf maturity, and climatic conditions such as temperature, humidity, rainfall, solar radiation, and seasonal variability, in addition to post-harvest handling, processing, and storage conditions (Wong et al. 2022). Freshly plucked tea leaves, on a dry weight basis, contain TPC comprising approximately 25–30% of their composition (Piyasena et al. 2023). These phenolic compounds are particularly rich in flavanols, flavonols, flavones, proanthocyanidins, flavandiols, and phenolic acids, while black tea additionally contains theaflavins and thearubigins, as well as other minor polyphenolic compounds (Piyasena, 2025). The characteristic color of black tea arises mainly from polyphenolic compounds formed during enzymatic oxidation, in which polyphenol oxidases catalyze the conversion of native polyphenols into key pigments, including orange–red theaflavins and dark brown thearubigins, as well as smaller amounts of theasinensins and theabrownins (Piyasena and Hettiarachchi 2025). Given the established health benefits of tea, TPC provides a quantifiable and integrative measure of the biochemical quality of tea, making it particularly relevant for nutraceutical applications and the production of specialty or functionally enriched teas. Therefore, TPC was selected as the primary parameter and quantified in this study.

Sri Lanka is characterized by diverse and seasonally variable climatic conditions that evidently influence the biochemical composition of tea leaves. These seasonal variations contribute to the manufacture of distinct regional teas, such as Uva and Dimbula teas, which are produced during specific periods and are renowned for their unique physicochemical and sensory characteristics (Kasuga et al. 2010). Thus, tea manufacturers often adjust oxidation and aeration durations during processing to enhance black tea quality and market value (Piyasena et al. 2023). The production of specialty teas tailored to particular climatic conditions is an emerging trend in Sri Lanka, leveraging seasonal variation to enhance flavor profiles and quality attributes. While previous research studies have examined seasonal variations in Sri Lankan black tea in antioxidant activity, caffeine, catechins, and TPC in fresh and processed teas during specific monsoon periods (Jayasekara et al. 2011; Jayasekara et al. 2014). Furthermore, biochemical constituents of Assam/Cambod and China type tea accessions evaluated in exotic germplasm were studied during both the wet and dry seasons (Kottawa-Arachchi et al. 2022). In vitro enzyme inhibitory activities, including lipase, α-amylase, xanthine oxidase, α-glucosidase, and acetylcholinesterase, as well as antioxidant activity, were systematically assessed in black tea and fresh tea leaves collected from fifteen distinct tea cultivars during the dry and wet seasons in Sri Lanka (Piyasena et al. 2025). Especially, most existing studies lack comprehensive long-term sampling strategies, relying instead on limited sample collections from the wet and dry seasons in Sri Lanka. The absence of consistent monthly sampling represents a significant knowledge gap in understanding seasonal variations in TPC in both fresh tea leaves and processed black tea. Besides, the study assessed the effects of freeze-drying (FD), oven-drying (OD; 60 °C), and conventional black tea (BT) processing on TPC retention to elucidate the impact of thermal and oxidative treatments on polyphenol stability. These methods were selected as representative techniques commonly used in tea research. FD served as a reference for maximal preservation of native TPC, OD at 60 °C represented controlled thermal dehydration with limited degradation and enzyme inactivation, and BT processing reflected industrial oxidative transformation. Collectively, these methods enabled a comparative evaluation of TPC retention under minimal, moderate, and intensive processing conditions relevant to nutraceutical and commercial tea production. Furthermore, this study investigated widely cultivated tea cultivars in Sri Lanka, among which DT-1 is recognized for producing high-quality black tea, whereas TRI 2025 is generally associated with the production of comparatively lower-quality black tea (Piyasena et al. 2022).

Therefore, the present study aimed to (i) investigate monthly fluctuations in TPC of two widely cultivated Sri Lankan tea cultivars, DT-1 and TRI 2025, in relation to climatic variables, including average temperature, total monthly rainfall, and average monthly sunshine hours; and (ii) evaluate the impact of different processing methods, including freeze-drying (FD) of fresh leaves, oven-drying (OD) at 60 °C, and conventional black tea (BT) processing, on TPC retention. The results of this study are expected to provide valuable insights into optimizing polyphenol retention for the production of polyphenol-enriched specialty teas and for the extraction of tea polyphenols for nutraceutical applications, while also informing strategies to maximize polyphenol content and

enhance the quality and functional value of Sri Lankan black teas.

## Materials and methods

### Chemicals and reagents

Folin–Ciocalteu's phenol reagent, gallic acid, methanol (purity 99%, HPLC grade), and sodium carbonate (anhydrous) were purchased from Sigma Aldrich, Germany.

### Sample collection

Tea leaf samples were collected over a 22-month period, from May 2019 to March 2021, at the St. Coombs Tea Estate in Talawakelle (6.9388° N, 80.6632° E), which is managed by the Tea Research Institute of Sri Lanka (TRISL). The accession numbers assigned to the respective cultivars, 2019/05/TRI 2025/01 and 2019/05/DT-1/01, have been deposited at the TRISL. Sampling was not conducted in August and September 2020 due to disruptions caused by the COVID-19 pandemic. The estate is located in the Central Province at an elevation of 1,368 m above sea level and features red-yellow podzolic soils, classified as Typic Hapludults (Piyasena et al. 2022). The region experiences a humid, subtropical-like climate with distinct dry and wet seasons. The dry season includes the months from January to March and from August to September with an average temperature of 28–30 °C, average rainfall of less than 100 mm, and sunshine duration of 7 to 8 h. The wet season includes months from April to July and from October to December, with an average temperature of 20–22 °C, average rainfall of more than 150 mm, and sunshine duration of 4.9 to 6.4 h (Piyasena et al. 2023). In the second week of each month, the harvestable shoots were collected from tea cultivars TRI 2025 and DT-1. Fresh leaves (100 g) from each tea cultivar were freeze-dried using a freeze dryer (Labconco FreeZone 4.5, USA) under vacuum conditions of 0.05 mbar at −40 °C for 24 h. Another 100 g of fresh leaf material was oven-dried at 60 °C for 24 h in a Memmert drying oven (UND300, Germany); humidity was not controlled and was assumed to be minimal due to the elevated drying temperature. The initial moisture content of fresh tea leaves ranged from 76 to 79%. An additional 100 g was used to miniature-process black tea at the Biochemistry Division of the TRISL.

Conditions of the Miniature Black Tea Manufacturing System (Teacraft, UK): Withering was carried out under a difference between wet and dry bulb temperatures of 4 °C, with a 75% relative humidity. Rolling was performed using an orthodox roller for 15 min, followed by maceration conducted twice in the roller. The macerated leaves were sieved using a mesh No. 8 (2.057 mm). Aeration/oxidation took place in an aeration cupboard, maintained at a dry bulb temperature of 25 °C, a wet bulb temperature of 25 °C, and 100% relative humidity. Drying was performed using a fluid bed dryer with an inlet temperature of 125 °C and an outlet temperature of 95 °C for a duration of 18–21 min under low-humidity hot air conditions conducive to rapid moisture removal. Finally, fiber removal was conducted using a fiber mat.

### Quantification of total polyphenol contents (TPC)

The dried tea leaves were ground into a fine powder, and an accurately weighed (0.200 ± 0.001 g) amount was transferred into extraction tubes (10 mL). Each sample was extracted with hot methanol (5 mL, 70% (v/v)) and vortexed (Lab-Line, USA). The tubes were then incubated (70 °C) for 10 min. Once cooled to room temperature, the samples were centrifuged (LAB-KITS, Korea) at 3,500 rpm for 10 min, and the supernatant was transferred into graduated tubes. This process was repeated, and the combined supernatants were made up to 10 mL (70% methanol), and 1 mL of this extract was further diluted to 100 mL (distilled water) in a volumetric flask and thoroughly mixed, and 1 mL of this extract was mixed with Folin–Ciocalteu's phenol reagent (5 mL of 10% (v/v)). Within 3–8 min, sodium carbonate solution (4 mL, 7.5% (w/v)) was added and incubated at room temperature for 1 h. The absorbance was measured at 765 nm using a UV-Vis spectrophotometer (Shimadzu, Japan). The calibration curve was developed using gallic acid (10 to 60 mg/L). TPC was given as a dry matter basis (w/w %) (ISO 14502-1:2005). Dry matter content was determined according to ISO 1573:1980. Precisely 5 g of tea samples were weighed and placed in pre-weighed aluminum moisture cans. The samples were then oven-dried (Memmert) at 103 ± 2 °C for 6 h until a constant weight was achieved. Dry matter content was calculated based on the difference in weight before and after drying (ISO 1573:1980).

### Analysis of moisture content

Tea samples weighing 5 g each were accurately weighed and transferred into aluminum moisture cans (pre-weighed). The samples were then dried (Memmert UND300, Germany) at a temperature of 103 ± 2 °C for 6 h, or until a constant weight was obtained, indicating complete moisture removal. The dry matter content was calculated in accordance with the ISO 1573:1980 standard (ISO 1573:1980).

### Meteorological data

Meteorological data, including average monthly temperature, total monthly rainfall, and monthly sunshine hours, were obtained from the Agronomy Division of the TRISL.

### Statistical analysis

Data analysis was performed by analysis of variance using SAS software (version 9.1). Treatment means were compared using Duncan's multiple range test at a significance level of $p \leq 0.05$. The results are presented as mean ± standard deviation (SD). Additionally, Pearson's correlation analysis was performed to assess the relationship between the TPC of tea cultivars and selected meteorological variables.

## Results

### Total polyphenol content (TPC) and climatic responses of the tea cultivar TRI 2025

Figure 1; Table 1 illustrate the influence of climatic variables on TPC in tea samples from the TRI 2025 cultivar across different processing methods: FD, OD, and BT samples ($p \leq 0.05$). Throughout the study period, OD samples generally showed higher TPC than BT, except in September 2019, December 2020, and January 2021, where differences were minimal and statistically insignificant (e.g., September 2019: OD 21.35 ± 0.05% vs. BT 21.90 ± 0.80%). The highest TPC in OD samples was observed in August 2019, while the highest in FD samples (27.55 ± 0.07%) occurred in February 2020, during the dry season, characterized by low rainfall (3.6 mm), high sunshine duration (7.9 h/day), and elevated temperatures (~ 19 °C). Conversely, the lowest TPC in FD (20.39 ± 0.58%) and BT (16.89 ± 1.01%) was recorded in November 2020, a wet month with 130.4 mm rainfall, 5.1 sunshine hours/day, and a mean temperature of 17 °C (see Table 1).

TPC in BT was recorded in July 2019 (23.25 ± 0.07%). Although July falls within the wet season with high rainfall (256.1 mm), the relatively warm temperature (18.4 °C) and moderate sunshine (~ 6 h/day) likely supported increased polyphenol accumulation. Similarly elevated TPC values in BT were observed in January 2020 (22.85 ± 0.49%), February 2020 (21.45 ± 0.21%), January 2021 (22.89 ± 0.39%), and February 2021 (22.81 ± 0.21%). OD samples, the highest TPC was recorded in August 2019 (27.70 ± 0.20%), followed by October 2019 (27.35 ± 0.05%), January 2020 (27.15 ± 0.35%), and February 2021 (26.96 ± 0.08%). The elevated TPC observed in January 2020 and February 2021 corresponds to warm, dry conditions. Interestingly, August and October 2019 also exhibited high TPC despite experiencing the highest rainfall (268.7 mm and 363.9 mm, respectively), low sunshine (~ 2 h/day), and cooler temperatures (17–18 °C), suggesting that stress-induced polyphenol biosynthesis triggered by excessive wetness and related plant stressors may have contributed to increased TPC. The lowest TPC in BT samples occurred in April 2020 (20.10 ± 0.20%) and December 2020. Based on these trends, the optimal months for harvesting tea leaves to maximize polyphenol content in this cultivar are January, February, July, August, and October.

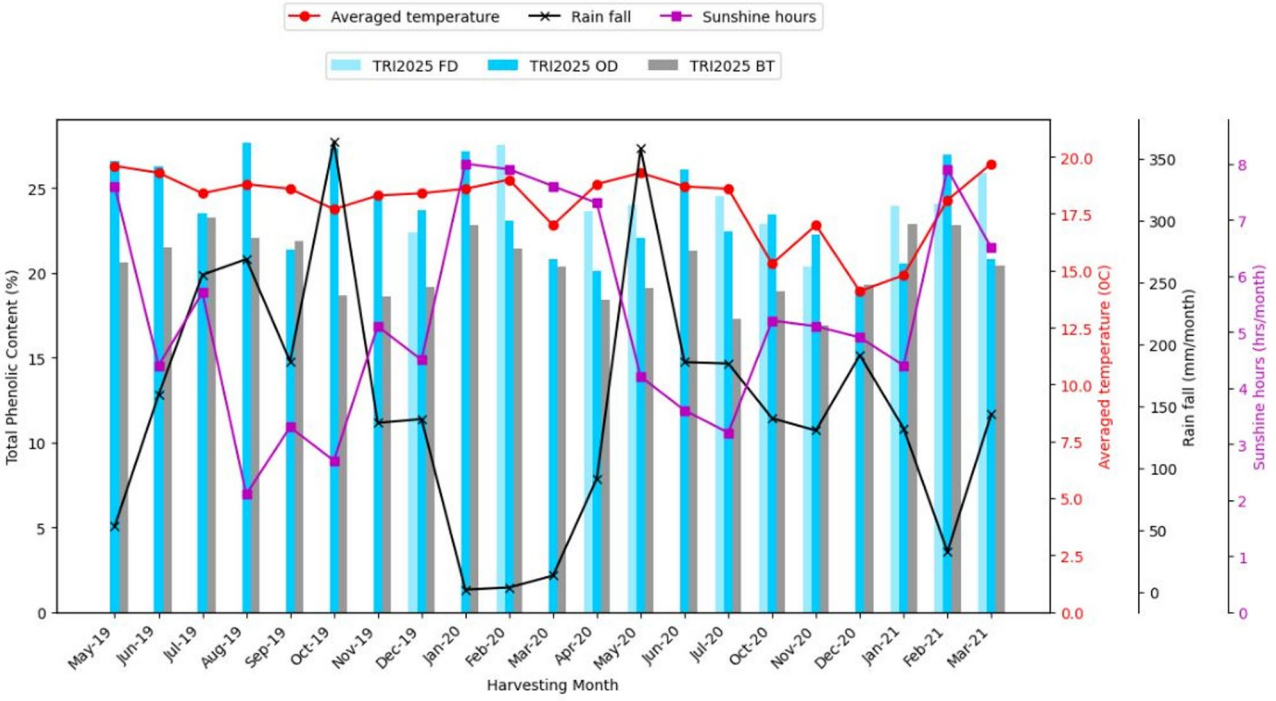



**Fig. 1** Climate data and TPC of FD, OD, and BT of cultivar TRI 2025 over the tested time period

**Table 1** Results of the mean separation of the TPC of FD, OD, and BT of tea cultivar TRI 2025 and cultivar DT-1 and the climate data of the respective months

| Harvesting month | TRI 2025 | | | DT-1 | | | Climate data | | |
|---|---|---|---|---|---|---|---|---|---|
| | FD | OD | BT | FD | OD | BT | Averaged temperature (°C) | Rainfall (mm/month) | Sunshine (number of hours/month) |
| 2019-May | – | $26.60^{abc} \pm 0.50$ | $20.60^{cd} \pm 0.14$ | – | $20.7^{fgh} \pm 0.71$ | $20.70^{a, b, c} \pm 0.28$ | 19.6 | 53.1 | 7.6 |
| 2019-Jun | – | $26.30^{bc} \pm 0.30$ | $21.50^{b} \pm 0.71$ | – | $23.40^{cde} \pm 0.21$ | $20.95^{b, c, d, e} \pm 0.35$ | 19.3 | 159.2 | 4.4 |
| 2019-Jul | – | $23.55^{def} \pm 0.20$ | $23.25^{a} \pm 0.07$ | – | $23.8^{cd} \pm 0.42$ | $19.95^{c, d, e} \pm 0.35$ | 18.4 | 256.1 | 5.7 |
| 2019-Aug | – | $27.70^{a} \pm 0.20$ | $22.05^{b} \pm 0.35$ | – | $25.95^{b} \pm 0.63$ | $18.4^{f} \pm 0.21$ | 18.8 | 268.7 | 2.1 |
| 2019-Sep | – | $21.35^{hi} \pm 0.05$ | $21.90^{b} \pm 0.28$ | – | $16.80^{j} \pm 0.28$ | $19.4^{c, d, e, f} \pm 0.28$ | 18.6 | 186.3 | 3.3 |
| 2019-Oct | – | $27.35^{ab} \pm 0.05$ | $18.70^{fg} \pm 0.28$ | – | $19.80^{ghi} \pm 0.11$ | $16.35^{g} \pm 0.63$ | 17.7 | 363.9 | 2.7 |
| 2019-Nov | – | $23.50^{d} \pm 0.00$ | $18.60^{fg} \pm 0.14$ | – | $19.60^{hi} \pm 0.42$ | $16.4^{g} \pm 0.14$ | 18.3 | 136.5 | 5.1 |
| 2019-Dec | $22.99^{fg} \pm 0.28$ | $23.10^{de} \pm 0.20$ | $19.15^{efg} \pm 0.70$ | $24.20^{cd} \pm 0.28$ | $23.55^{cd} \pm 0.21$ | $18.65^{e, f} \pm 0.23$ | 18.4 | 139.6 | 4.5 |
| 2020-Jan | – | $27.15^{abc} \pm 0.35$ | $22.85^{a} \pm 0.49$ | | $27.3^{a} \pm 0.14$ | $21.60^{a} \pm 0.25$ | 18.6 | 1.9 | 8.0 |
| 2020-Feb | $27.55^{a} \pm 0.07$ | $23.05^{efg} \pm 0.25$ | $21.45^{ab} \pm 0.21$ | $27.20^{a} \pm 0.56$ | $25.55^{b} \pm 0.75$ | $21.35^{a, b} \pm 0.07$ | 19.0 | 3.6 | 7.9 |
| 2020-Mar | – | $20.81^{ij} \pm 0.06$ | $20.40^{d} \pm 0.21$ | – | $20.81^{f, g} \pm 0.32$ | $20.17^{b, c, d} \pm 0.36$ | 17.0 | 13.21 | 7.6 |
| 2020-Apr | $23.63^{de} \pm 0.08$ | $20.10^{jk} \pm 0.20$ | $18.45^{g} \pm 0.14$ | $22.77^{e} \pm 0.07$ | $19.05^{i} \pm 0.63$ | $19.15^{d, e, f} \pm 1.76$ | 18.8 | 91.6 | 7.3 |
| 2020-May | $24.00^{cd} \pm 0.01$ | $22.10^{gh} \pm 0.10$ | $19.10^{efg} \pm 0.14$ | $23.48^{de} \pm 0.01$ | $22.70^{e} \pm 0.89$ | $18.25^{f} \pm 0.28$ | 19.3 | 358.0 | 4.2 |
| 2020-Jun | – | $26.10^{c} \pm 1.20$ | $21.30^{bc} \pm 0.14$ | – | $24.45^{c} \pm 0.78$ | $18.4^{f} \pm 1.13$ | 18.7 | 185.6 | 3.6 |
| 2020-Jul | $24.50^{d} \pm 0.72$ | $22.47^{fgh} \pm 0.43$ | $17.30^{h} \pm 0.10$ | $22.78^{e} \pm 0.20$ | $20.35^{fgh} \pm 0.23$ | $16.8^{g} \pm 0.48$ | 18.6 | 184.5 | 3.2 |
| 2020-Oct | $23.99^{de} \pm 0.28$ | $23.00^{def} \pm 0.41$ | $18.91^{efg} \pm 0.07$ | $23.37^{de} \pm 0.42$ | $22.37^{e} \pm 0.52$ | $14.58^{h} \pm 032$ | 15.3 | 140.2 | 5.2 |
| 2020-Nov | $21.99^{h} \pm 0.58$ | $22.00^{gh} \pm 0.07$ | $16.89^{h} \pm 1.01$ | $23.97^{de} \pm 0.28$ | $22.93^{de} \pm 1.07$ | $16.71^{g} \pm 0.01$ | 17.0 | 130.4 | 5.1 |
| 2020-Dec | – | $18.15^{k} \pm 0.11$ | $19.29^{ef} \pm 0.09$ | – | $22.37^{e} \pm 0.28$ | $18.23^{f} \pm 0.09$ | 14.1 | 191.2 | 4.9 |
| 2021-Jan | $23.99^{de} \pm 0.06$ | $20.58^{ij} \pm 0.14$ | $22.89^{a} \pm 0.39$ | $23.40^{de} \pm 0.56$ | $20.99^{f} \pm 0.25$ | $21.34^{a, b} \pm 1.15$ | 14.8 | 132.1 | 4.4 |
| 21-Feb | $24.09^{c, d} \pm 0.16$ | $26.96^{abc} \pm 0.08$ | $22.81^{a} \pm 0.21$ | $25.04^{b, c} \pm 0.16$ | $25.94^{b} \pm 0.08$ | $22.07^{a} \pm 092$ | 18.1 | 32.6 | 7.9 |
| 21-Mar | $25.94^{b} \pm 0.05$ | $20.81^{ijd} \pm 0.09$ | $20.41^{d} \pm 0.23$ | $25.47^{b} \pm 0.72$ | $20.81^{fgd} \pm 0.02$ | $20.17^{b, c, d} \pm 0.24$ | 19.7 | 143.5 | 6.5 |

The values in the table are given as mean± SD of triplicates. The means, that do not share the same subscription letter (within a column), are significantly different at 95% confident (α=0.05). *TPC* Total polyphenol content, *OD* Oven Dried, *BT* Black Tea

### Total polyphenol content (TPC) and climatic responses of the tea cultivar DT-1

Figure 2 illustrates the influence of climatic factors on the TPC of DT-1 tea samples. Overall, OD samples generally exhibited higher TPC than BT, except in September 2019, April 2020, and January 2021. In April 2020 (OD: 19.05 ± 0.63% vs. BT: 19.15 ± 0.76%) and January 2021 (OD: 20.99 ± 0.25% vs. BT: 21.34 ± 1.15%), the differences were minimal and not statistically significant. However, in September 2019, BT samples exhibited a notably higher TPC than OD samples (BT: 19.40 ± 0.28% vs. OD: 16.80 ± 0.28%). The highest TPC in OD samples was observed in January 2020, while the peak TPC in freeze-dried (FD) samples occurred in February 2020 (27.20 ± 0.56%), likely reflecting warm, dry conditions. The lowest TPC in FD samples (22.78 ± 0.20%) was

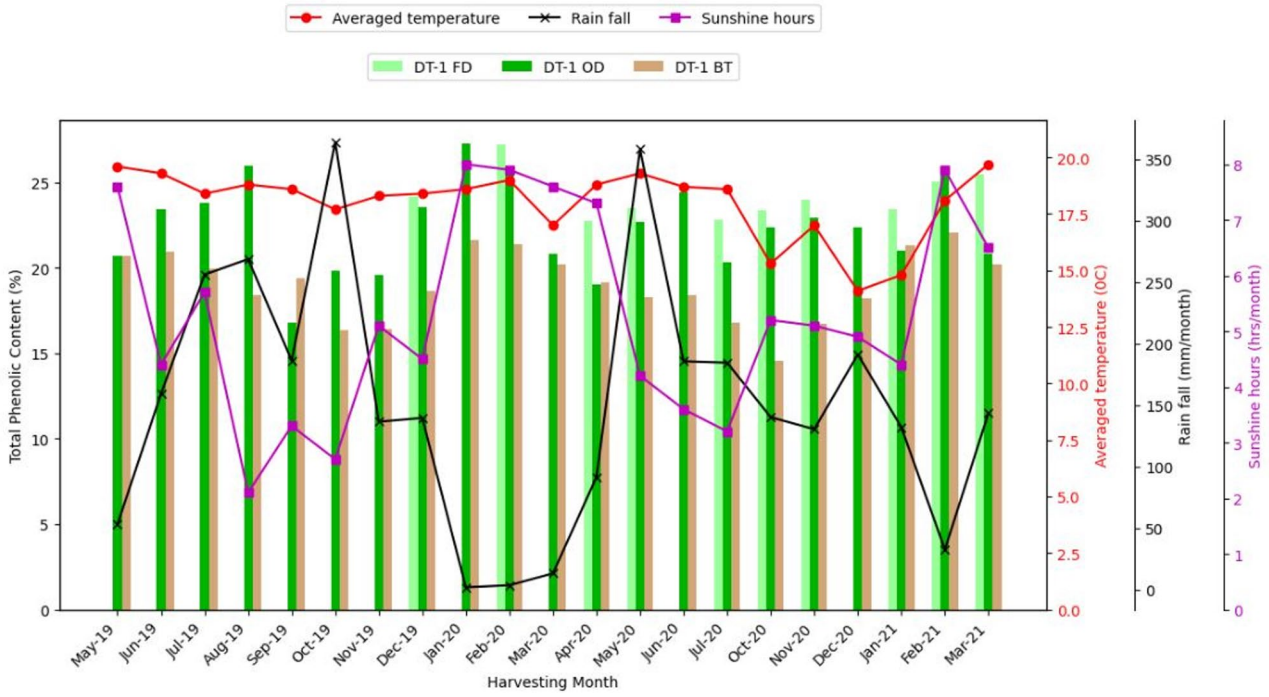



**Fig. 2** Climate data and TPC of FD, OD, and BT of cultivar DT-1 over the tested time period

recorded in July 2020, potentially due to reduced sunshine (3.2 h/day), despite favorable temperature (18.6 °C) and rainfall (184.5 mm/month) conditions.

In this study, the highest total polyphenol content (TPC) in oven-dried (OD) samples of the DT-1 cultivar was recorded in January 2020 (27.3 ± 0.14%), closely followed by the peak in freeze-dried (FD) samples in February 2020. Both months were characterized by warm, dry conditions with low rainfall and extended sunshine, which are favorable for polyphenol accumulation. The lowest TPC in OD samples (16.80 ± 0.28%) occurred in September 2019, the lowest across both cultivars, under conditions of high rainfall (186.3 mm/month), limited sunshine (3.3 h/day), and an average temperature of 18.6 °C, likely reflecting the cumulative effects of prolonged wet conditions. For black tea (BT), the highest TPC was observed in February 2021 (22.07 ± 0.92%), also coinciding with warm, dry weather, whereas the lowest TPC in BT samples (14.58 ± 0.32%) occurred in October 2020, when the average temperature dropped to 15.2 °C, one of the lowest recorded during the study, negatively affecting polyphenol levels. FD samples similarly showed reduced TPC during this period. Although January typically corresponds to the wet season, January 2021 experienced the lowest rainfall recorded in the study, further supporting January and February as the optimal harvest months for maximizing TPC in the DT-1 cultivar.

### Correlation analysis of TPC in oven-dried (OD) and black tea (BT) samples with meteorological parameters

Table 2 presents the Pearson's correlation coefficients (r) between TPC in OD and BT samples and selected meteorological parameters. A strong, significant positive correlation was observed between OD samples of the two cultivars ($r = +0.890$, $p \le 0.05$) and between BT samples of the two cultivars ($r = +0.750$, $p \le 0.05$), indicating consistent TPC variation between cultivars during processing. However, no significant correlation ($p \ge 0.05$) was detected between OD and BT samples within the same cultivar. For the TRI 2025 cultivar, TPC in OD samples exhibited a significant positive correlation with average temperature ($r = +0.501$, $p \le 0.05$), whereas for DT-1, TPC in OD samples was positively correlated with sunshine duration ($p \le 0.05$). TPC in BT samples of TRI 2025 showed no significant correlation with climatic factors ($p \ge 0.05$), while TPC in BT samples of DT-1 displayed notable correlations with rainfall ($r = -0.543$, $p \le 0.05$) and sunshine duration ($r = +0.601$, $p \le 0.05$). Overall, these findings indicate that although specific climatic parameters may influence TPC under certain conditions, no consistent pattern is evident across cultivars and processing methods, suggesting that the relationship between climate and TPC in processed teas is complex and cultivar-specific.

**Table 2** Pearson's linear correlation coefficients (*r*) within TPC of OD and BT of tea cultivars, averaged temperature, rainfall and sunshine hours

| TRI 2025 OD | TRI 2025 OD | TRI 2025 BT | DT-1 OD | DT-1 BT | Averaged temperature | Rain fall |
|---|---|---|---|---|---|---|
| TRI 2025 BT | 0.215<br>$p = 0.337$ | | | | | |
| DT-1 OD | 0.890<br>$p$=0.000 | 0.274<br>$p = 0.217$ | | | | |
| DT-1 BT | 0.266<br>$p = 0.232$ | 0.750<br>$p = 0.000$ | 0.382<br>$p = 0.079$ | | | |
| Averaged Temperature | 0.501<br>$p = 0.018$ | 0.140<br>$p = 0.536$ | 0.357<br>$p = 0.103$ | 0.348<br>$p = 0.113$ | | |
| Rain fall | − 0.192<br>$p = 0.391$ | − 0.223<br>$p = 0.319$ | − 0.310<br>$p = 0.160$ | − 0.543<br>$p = 0.009$ | 0.081<br>$p = 0.721$ | |
| Sunshine hours | 0.368<br>$p = 0.092$ | 0.228<br>$p = 0.307$ | 0.499<br>$p = 0.018$ | 0.601<br>$p = 0.003$ | 0.255<br>$p = 0.253$ | 0.808<br>$p = 0.000$ |

**Table 3** Pearson's linear correlation coefficients (*r*) within TPC of freeze-dried leaves of tea cultivars, averaged temperature, rainfall and sunshine

| TRI 2025 FD | TRI 2025 FD | DT-1 FD | TRI2025 OD | TRI2025 BT | DT-1 OD | DT-1 BT | Averaged temperature | Rainfall |
|---|---|---|---|---|---|---|---|---|
| DT-1 FD | 0.613<br>$p = 0.060$ | | | | | | | |
| TRI 2025 OD | −0.059<br>$p = 0.871$ | 0.317<br>$p = 0.372$ | | | | | | |
| TRI 2025 BT | 0.520<br>$p = 0.124$ | 0.501<br>$p = 0.140$ | 0.271<br>$p = 0.450$ | | | | | |
| DT-1 OD | 0.128<br>$p = 0.724$ | 0.686<br>$p = 0.029$ | 0.816<br>$p = 0.004$ | 0.422<br>$p = 0.224$ | | | | |
| DT-1 BT | 0.559<br>$p = 0.093$ | 0.562<br>$p = 0.091$ | 0.104<br>$p = 0.776$ | 0.826<br>$p = 0.03$ | 0.322<br>$p = 0.364$ | | | |
| Averaged temperature | 0.443<br>$p = 0.199$ | 0.354<br>$p = 0.315$ | − 0.003<br>$p = 0.994$ | − 0.165<br>$p = 0.649$ | 0.060<br>$p = 0.870$ | 0.253<br>$p = 0.480$ | | |
| Rainfall | − 0.250<br>$p = 0.486$ | − 0.537<br>$p = 0.109$ | 0.302<br>$p = 0.396$ | − 0.411<br>$p = 0.238$ | − 0.354<br>$p = 0.316$ | − 0.444<br>$p = 0.199$ | 0.115<br>$p = 0.753$ | |
| Sunshine hours | 0.425<br>$p = 0.221$ | 0.643<br>$p = 0.045$ | 0.245<br>$p = 0.495$ | 0.467<br>$p = 0.173$ | 0.389<br>$p = 0.266$ | 0.571<br>$p = 0.085$ | 0.282<br>$p = 0.430$ | − 0.723<br>$p = 0.018$ |

### Correlation analysis of TPC in freeze dried (FD) samples with meteorological parameters

Table 3 presents the Pearson's correlation coefficients (r) between total polyphenol content (TPC) in freeze-dried (FD) and black tea (BT) samples and selected climatic variables over a ten-month period. For the TRI-2025 cultivar, TPC in FD samples showed a moderate, non-significant positive correlation with TPC in BT samples ($r = +0.520$, $p \geq 0.05$), and similarly non-significant correlations with FD ($r = +0.613$) and BT ($r = +0.559$) samples of the DT-1 cultivar. In contrast, TPC in FD samples of DT-1 exhibited a significant positive correlation with its oven-dried (OD) samples ($r = +0.686$, $p \leq 0.05$) and a strong but non-significant correlation with its BT samples ($p \geq 0.05$). No significant correlations were observed between TPC in FD samples and either average temperature or rainfall for either cultivar. However, in DT-1, TPC in FD samples exhibited a notable positive correlation with sunshine duration, suggesting that sunlight may play a key role in polyphenol accumulation in fresh leaves of this cultivar (see Table 3).

## Discussion

This study investigated variations in total polyphenol content (TPC) in oven-dried (OD) tea leaves and black tea (BT) over a continuous period of approximately two years. In addition, the TPC of freeze-dried (FD) tea leaves was quantified over a 10-month period, with samples collected monthly. Tea samples from two distinct cultivars, TRI 2025 and DT-1, were collected from the same geographical location to control for site-specific environmental variability. Climatic parameters reported include the temperatures recorded on the day of sample collection, along with average monthly rainfall and sunshine duration. TPC was quantified in black tea, oven-dried tea leaves dried at 60 °C, and freeze-dried tea leaves. Freeze-dried tea leaves were used to assess the polyphenol content of raw tea leaves, providing a baseline for understanding natural variations. Black tea samples were analyzed to evaluate how polyphenol content changes with climatic conditions, offering insight into the production of seasonal or specialty teas. According to the literature, conventional extraction methods generally

yield maximum TPC at temperatures between 60 and 80 °C (Antony and Farid 2020). In contrast, total flavonoid content has been reported to decrease at temperatures above 60 °C in flax and canola seed cake (Mokrani and Madani 2016). Additionally, within the temperature range of 20–60 °C, an initial increase in temperature has been shown to enhance anthocyanin extraction (Cacace and Mazza 2003). Based on these findings, 60 °C was selected as the drying temperature in the present study to minimize thermal degradation while enabling accurate quantification of TPC in tea leaves.

According to the results of this study, BT exhibited the lowest TPC compared to both OD and FD samples, reflecting the degree of oxidation involved, with black tea showing the lowest polyphenol retention due to full enzymatic oxidation during processing (Aaqil et al. 2023; Piyasena et al. 2025). TPC was notably higher during the warm and dry months of January and February, followed by the pre-monsoon months of June and July. Conversely, the lowest TPC values were recorded during the wet season months of October, November, and December, which coincide with the monsoon period. While the TRI 2025 cultivar generally exhibited the highest TPC across all three sample types, the DT-1 cultivar produced closely comparable results, indicating minimal variation between the two cultivars in their response to weather conditions. It is reported that the optimal ambient temperature range for high-yielding Sri Lankan tea cultivars is about 22 °C (Wijeratne et al. 2007). The elevated TPC observed during warm, dry months in both study years is consistent with this range and may be attributed to drought-induced stress. Such stress conditions are known to trigger polyphenol biosynthesis as a protective response (Lv et al. 2021). Given the ongoing rise in global temperatures and altered precipitation patterns, the impact of climatic variability on tea quality has become important (Wijeratne et al. 2007).

Several studies have documented seasonal fluctuations in TPC and related biochemical parameters. Significantly elevated TPC and antioxidant activity are reported during the spring season (March–May) in Nagaland, India (Jamir et al. 2020). Deka et al. (2023) investigated seasonal variation in catechin and caffeine content across pre-monsoon, monsoon, and autumn harvests. The monsoon season produced the highest levels of total catechins, epigallocatechin gallate (EGCG), epicatechin gallate (ECG), and caffeine. EGCG was found to be the most abundant catechin in all seasons. Seasonal fluctuations in catechin and caffeine levels were attributed to environmental factors such as day length, sunlight, and temperature (Deka et al. 2023). Govindasamy et al. (2023) identified increased levels of polyphenols, catechins, antioxidant activity, and essential nutrients (Ca, N, P, Mn, Zn, Fe) during the summer months (March–June) in Tamil Nadu, India (Govindasamy et al. 2023). The concentrations of ECG, EGCG, and total catechins in fresh tea shoots from North Queensland, Australia, were significantly higher during the warmer months (April–May) and decreased during cooler months (July–September) (Yao et al. 2025). Paiva et al. (2021) compared the antioxidant activity, total phenolic and flavonoid contents, catechin composition, and caffeine levels of Azorean green tea harvested across different seasons over two consecutive years in Portugal. Antioxidant activity was generally higher during summer (July–August). Seasonal and interannual variations in total phenolic and flavonoid contents reflected the influence of climatic conditions (Paiva et al. 2021). The seasonal variation in TPC in fresh tea shoots was collected from several tea clones grown in the Eastern Black Sea region of Turkey across three harvest periods over two consecutive years. TPC was generally lower during the cooler early harvests and increased in the later, warmer harvests (Erturk et al. 2010).

The findings of this study are consistent with findings reported in the previous research. TPC was significantly higher during the warm and dry months, whereas the lowest TPC values were recorded during the wet monsoon season. Although the tea cultivar TRI 2025 consistently exhibited the highest TPC across all three sample types, the values observed in the DT-1 cultivar were closely comparable, suggesting minimal variation between the two cultivars in their phenolic response to seasonal weather conditions. The study also demonstrated that leaf processing methods significantly influence the retention of phenolic compounds. Freeze-drying preserved the highest concentrations of TPC, followed by oven drying at 60 °C and black tea processing. This indicates that minimal thermal degradation and oxidation during processing are critical for maintaining the integrity of bioactive constituents. These findings underscore the importance of both harvest timing and post-harvest handling practices in optimizing tea quality and maximizing associated health benefits. Therefore, harvesting tea leaves during warm, dry periods may enhance the polyphenol content of raw material, making it particularly suitable for the production of high-value, functionally enriched, and specialty tea products.

This study has several limitations, including the relatively short duration of two years, which may not adequately capture long-term climatic variability and its effects on tea polyphenol dynamics. Although general environmental parameters were considered, direct correlations with specific factors such as soil moisture, relative humidity, and solar radiation were not evaluated. In addition to TPC, the quantification of individual polyphenolic compounds such as theaflavins, catechins, thearubigins, anthocyanins, phenolic acids, amino acids, etc. may provide more detailed insights into tea quality. Moreover, the

influence of biotic factors, including plant pest incidence, stress responses, and agronomic practices, should be explored, as these may significantly impact TPC biosynthesis. Future research should examine the influence of both abiotic and biotic factors, together with the chemical profile of tea, and evaluate their direct relationships with polyphenol accumulation. Furthermore, extending the study period beyond two years would allow for a more comprehensive assessment of the effects of long-term climatic variability on tea chemistry.

## Conclusions

This study reports the monthly variations in total phenolic content (TPC) of two tea cultivars in Sri Lanka over two consecutive years. The findings demonstrate that TPC in both cultivars varied significantly in response to prevailing weather conditions during the study period. These fluctuations suggest that environmental factors such as temperature, rainfall, and sunlight exposure significantly influence polyphenol biosynthesis or secondary metabolite production. Among the different processing methods, freeze-dried tea leaves consistently showed the highest TPC, followed by oven-dried tea leaves and black tea, regardless of cultivar. Furthermore, tea leaves harvested during dry months were shown to be more appropriate for producing high-quality black tea in both tea cultivars. Nevertheless, further research is warranted that comprehensively considers both abiotic and biotic factors to elucidate their specific roles and potential synergistic effects on the (TPC) of tea cultivars.

### Author contributions

K. G. Nelum P. Piyasena- Conceptualization, Funding acquisition, Investigation; Methodology; Project administration; Resources; Supervision; Writing - original draft; and Writing - review & editing, Jayani Kalinga- Data analysis, Writing - original draft; and Writing - review & editing, W. S. Jayasinghe- Data curation; Formal analysis, M. A. B. Ranatunga- Conceptualization.

### Funding

Not applicable.

### Data availability

No datasets were generated or analysed during the current study.

## Declarations

### Ethics approval and consent to participate

The collection of the leaves of *Camellia sinensis* L. O. (Kuntze) complied with local or national guidelines. Leaves were collected in the St. Coombs Tea Estate in Talawakelle (6.9388° N, 80.6632° E), which is managed by the Tea Research Institute of Sri Lanka. As tea is a commercially grown plant, a special license or special permission is not required to collect the tea leaves. This review did not require ethical approval or consent for participation.

### Consent for publication

Not applicable.

### Competing interests

The authors declare no competing interests.

Received: 20 November 2025 / Accepted: 17 February 2026

Published online: 26 February 2026

### References

Aaqil M, Peng C, Kamal A, Nawaz T, Zhang F, Gong J (2023) Tea harvesting and processing techniques and its effect on phytochemical profile and final quality of black tea: a review. Foods 12:4467. https://doi.org/10.3390/foods12244467

Ahmed S, Griffin TS, Kraner D, Schaffner MK, Sharma D, Hazel M, Leitch AR, Orians CM, Han W, Stepp JR, Robbat A (2019) Environmental factors variably impact tea secondary metabolites in the context of climate change. Front Plant Sci 10:939. https://doi.org/10.3389/fpls.2019.00939

Antony A, Farid M (2020) Effect of temperatures on polyphenols during extraction. Appl Sci 12(4):2107. https://doi.org/10.3390/app12042107

Basnayake P, Piyasena KN, Amarakoon AT, Ranatunga MA (2025) Evaluation of the anthocyanin content in Sri Lankan tea cultivars. Discover Plants 2:178. https://doi.org/10.1007/s44372-025-00267-4

Cacace JE, Mazza G (2003) Optimization of extraction of anthocyanins from black currants with aqueous ethanol. J Food Sci 68:240–248. https://doi.org/10.1111/j.1365-2621.2003.tb14146.x

Deka H, Sarmah PP, Chowdhury P, Rajkhowa K, Sabhapondit S, Panja S, Karak T (2023) Impact of the season on total polyphenol and antioxidant properties of tea cultivars of industrial importance in Northeast India. Foods 12:3196. https://doi.org/10.3390/foods12173196

Erturk Y, Ercisli S, Sengul M, Eser Z, Haznedar A, Turan M (2010) Seasonal variation of total phenolic, antioxidant activity and minerals in fresh tea shoots (*Camellia sinensis* var. *sinensis*). Pak J Pharm Sci 23:69–74

Govindasamy K, Sugumar DA, Kandan NM, Nagaprasad N, Ramaswamy K (2023) Seasonal variations in the phenolic profile, antioxidant activity, and mineral content of south Indian black tea (*Camellia sinensis* (L.) O. Kuntze). Sci Rep. https://doi.org/10.1038/s41598-023-45711-1

ISO (1980) ISO 1573: Tea—Determination of loss in mass at 103°C

ISO (2005) ISO 14502-1: Determination of substances characteristic of green and black tea—Part 1: Content of total polyphenols—Folin–Ciocalteu method

Jamir T, T A (2020) Seasonal variations in antioxidant capacities and phenolic contents of tea leaf extracts. Asian J Pharm Clin Res 13:108–112

Jayasekera S, Molan AL, Garg M, Moughan PJ (2011) Variation in antioxidant potential and total polyphenol content of fresh and fully-fermented Sri Lankan tea. Food Chem 125:536–541. https://doi.org/10.1016/j.foodchem.2010.09.045

Jayasekera S, Kaur L, Molan AL, Garg ML, Moughan PJ (2014) Effects of season and plantation on phenolic content of unfermented and fermented Sri Lankan tea. Food Chem. https://doi.org/10.1016/j.foodchem.2013.12.005

Kasuga H, Yamazaki Y, Baba A, Okajima S (2010) Characteristic odorants of Sri Lankan black teas: Uva, Nuwara Eliya, Dimbula, Kandy, and Ruhuna varietals. In: Flavors in noncarbonated beverages. ACS. pp 21–31

Kottawa-Arachchi JD, Ranatunga MA, Amarakoon AM, Gunasekare MT, Attanayake RN, Sharma RK, Chaudhary HK, Sood VK, Katoch R, Banyal DK, Piyasena KG (2022) Variation of catechin and caffeine content in exotic tea collection in Sri Lanka. Food Chem Adv. https://doi.org/10.1016/j.focha.2022.100108

Lv Z, Zhang C, Shao C, Liu B, Liu E, Yuan D, Zhou Y, Shen C (2021) Research progress on the response of tea catechins to drought stress. J Sci Food Agric. https://doi.org/10.1002/jsfa.11330

Mokrani A, Madani K (2016) Effect of solvent, time and temperature on the extraction of phenolic compounds and antioxidant capacity of peach (*Prunus persica* L.) fruit. Sep Purif Technol 162:68–76

Nelum KG, Piyasena P, Ranatunga MA, Jayawardhane SA, Edirisinghe EN, Tharangika HB, Ghouse AS, Abayarathne AA, Jayasinghe WS, Abeysinghe IS, Hettiarachchi LS (2023) Prediction of glucose and sucrose values of black tea using NIR spectroscopy and chemometrics. Food Humanity. https://doi.org/10.1016/j.foohum.2023.10.016

Paiva L, Lima E, Motta M, Marcone M, Baptista J (2021) Influence of seasonal and yearly variation on phenolic profiles, caffeine, and antioxidant activities of green tea from Azores. Appl Sci. https://doi.org/10.3390/app11167439

Piyasena NP (2025) Milk in tea: exploring the chemistry and biological activities. Food Sci Biotechnol. https://doi.org/10.1007/s10068-025-01841-y

Piyasena KN, Hettiarachchi LS (2025) Identification of key flavor compounds and color substances in tea: a review. Discover Food 5:250. https://doi.org/10.1007/s44187-025-00545-w

Piyasena KN, Hettiarachchi LS, Jayawardhane SA, Edirisinghe EN, Jayasinghe WS (2022) Evaluation of inherent fructose, glucose and sucrose concentrations in tea leaves (Camellia sinensis L.) and in black tea. Appl Food Res. https://doi.org/10.1016/j.afres.2022.100100

Piyasena KN, Abayarathne AA, Weerakoon NC, Edirisinghe EN, Jayasinghe WS, Napagoda MT, Hettiarachchi LS, Abeysinghe IS (2023) Chemical and biological characteristics of Sri Lankan white tea. Food Humanity. https://doi.org/10.1016/j.foohum.2023.08.019

Piyasena KN, Napagoda MT, Kalinga J, Abayarathne AA, Jayawardhane PA, Siriwardhane KU, Ranatunga MA, Jayasinghe L (2024) Variation of elastase, collagenase, tyrosinase inhibitory and antioxidant potential of different tea cultivars. Discover Chem 1:36. https://doi.org/10.1007/s44371-024-00041-7

Piyasena KN, Ranatunga MA, Napagoda MT, Amarasinghe NR, Abayarathne AA, Jayasinghe L (2025) Evaluation of antioxidant and digestive enzyme inhibitory activities of Sri Lankan tea cultivars. Discover Plants. https://doi.org/10.1007/s44372-025-00122-6

Wijeratne MA, Anandacoomaraswamy A, Amarathunga MK, Ratnasiri J, Basnayake BR, Kalra N (2007) Assessment of impact of climate change on productivity of tea plantations in Sri Lanka. J Natl Sci Foundation Sri Lanka 35(2):119–126

Wong M, Sirisena S, Ng K (2022) Phytochemical profile of differently processed tea: a review. J Food Sci 87:1925–1942. https://doi.org/10.1111/1750-3841.16137

Yao L, Caffin N, D'arcy B, Jiang Y, Shi J, Singanusong R, Liu X, Datta N, Kakuda Y, Xu Y (2025) Seasonal variations of phenolic compounds in Australia-grown tea (Camellia sinensis). J Agric Food Chem. https://doi.org/10.1021/jf050382y

## Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.